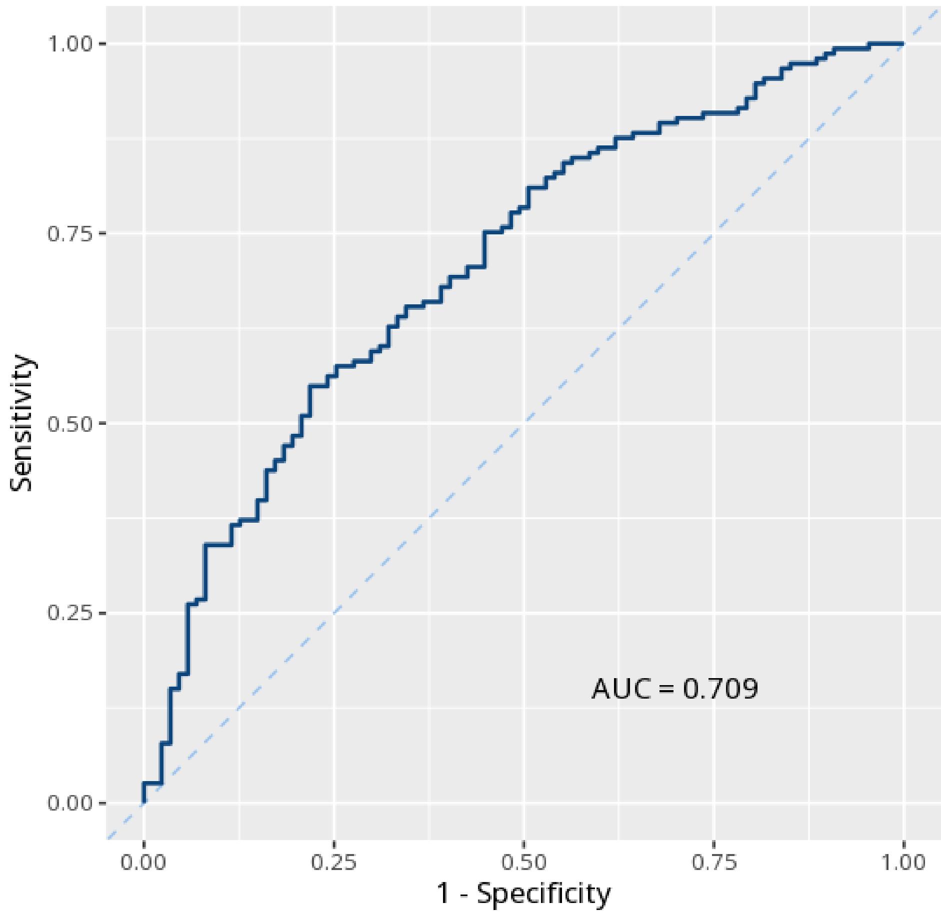
1 Logistic regression

	Log-odds (B)	Odds ratio (OR)
(Intercept)	−2.69	0.07
	(0.96)	[< 0.01, 0.43]
pretest_score	0.04	1.04
	(0.02)	[1.01, 1.08]
study_hours_per_week	0.19	1.21
	(0.06)	[1.08, 1.37]
teaching_method: Flipped	0.96	2.61
	(0.37)	[1.27, 5.52]
teaching_method: Lecture	−0.50	0.60
	(0.33)	[0.31, 1.16]
Num.Obs.	240	
Nagelkerke R^2	0.17	
Omnibus χ^2	31.41 (p <.001)	
Log.Lik.	−141.46	
AIC	292.91	
BIC	310.31	
Accuracy	69.2%	
Sensitivity	86.9%	
Specificity	37.9%	
AUC [95% CI]	.71 [.64, .78]	

Predicted \ Observed	No	Yes	% correct
No	33	20	62.3%
Yes	54	133	71.1%



Each predictor's odds ratio tells how the odds of the outcome change per unit (>1 increases, <1 decreases); a term carries weight when its OR confidence interval excludes 1, and the inline omnibus χ^2 (with its p) tests the model as a whole. Model fit and the ROC/AUC show how well it classifies overall, alongside accuracy (overall % correct), sensitivity (% of actual events correctly predicted), and specificity (% of actual non-events correctly predicted). Your significance threshold (α) is 0.05.

Predictor pretest_score was associated with the outcome, OR=1.04, 95% CI [1.01, 1.08], p = .025 (AUC=.71).

Statistical basis: Logistic regression coefficients (log-odds / odds ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Nagelkerke pseudo-R^2 (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.); ROC curve / AUC

(Robin X, Turck N, Hainard A, et al. (2011). "pROC: an open-source package for R and S+ to analyze and compare ROC curves." BMC Bioinformatics, 12, 77.).